

Capacity & Cost Analysis – World Class URL Shortener

1. Traffic Modeling

Workload Profile:

- Read-heavy: Redirects vastly outnumber short-URL creations ($\geq 50:1$ ratio).
- Global distribution: Majority of reads at edges near users; writes funneled into regional control planes.
- Hot vs cold keys:
 - * Hot set: Top 1–5% of codes handle 80–90% of traffic.
 - * Cold set: Long tail of rarely accessed codes still must be served.

Traffic Estimates (per busy region):

- Redirect QPS: 200K (p95 <50ms)
- Create/update QPS: 10K
- Analytics events: ~210K/sec
- Writes to store: Links = 10K/sec; Click events batched

2. Data Volume & Storage

Link Metadata Store:

- Avg link record: ~300 bytes
- Annual new links: 10K/sec $\rightarrow$ ~315B/year $\rightarrow$ ~95GB raw/year
- With replication ($\times 3$) & indexes: ~300GB/year

Click Analytics Store:

- Raw event: ~200 bytes compressed
- 200K/sec $\rightarrow$ ~6.3T/year $\rightarrow$ ~1.26TB/year compressed
- Retention: Hot (90 days in ClickHouse/BigQuery), Cold (S3/GCS indefinitely)

3. Cache Layer Sizing

Redis (Regional):

- Holds hot working set (e.g., top 1% of active keys)
- Key size ~ 60 bytes; value ~ 300 bytes $\rightarrow$ ~360 bytes total
- Hot set: 1% of 1B keys = 10M keys $\rightarrow$ ~3.6GB data; $\times 2$ –3 overhead $\rightarrow$ ~10GB/shard
- Shards: 12–24 for redundancy & performance

Edge Cache:

- Top few million hottest keys
- Refresh via change streams

4. Network Capacity

Redirect path:

- Edge hit: ~0.5KB in/out (302 headers)
- Cache miss: +1KB Redis traffic
- 200K QPS $\rightarrow$ ~800Mbps–1Gbps per busy region

Analytics ingestion:

- Kafka brokers handle ~210K events/sec $\rightarrow$ ~40MB/sec ingest

5. Compute Sizing

- Edge Workers: CPU-bound on TLS + redirect headers (~10K RPS/core)
- Regional API nodes: Handle writes, misses, admin calls
- Stream processors: Enrich & aggregate analytics

6. Cost Analysis (per busy region)

CDN/Edge compute: \$0.05 per 1M requests → ~\$2K–\$4K
Redis Cluster: ~\$600/node/mo, 12–24 shards → ~\$7K–\$14K
Durable store (DynamoDB): \$5K–\$15K
Kafka/Flink: ~\$6K–\$18K
ClickHouse/BigQuery: ~\$1K–\$5K
S3/GCS Archive: ~\$200–\$500
Total: \$20K–\$80K per month

7. Scaling Strategies

- Shard-by-tenant/domain to localize hot keys
- Promote hot keys to edge
- Adaptive throughput provisioning
- Batch writes for analytics
- Offload cold data to cheap storage